Meta Ad Performance Dataset

Dataset Description

This dataset contains comprehensive Meta (Facebook/Instagram) advertising performance data, structured to mirror real-world ad platform analytics. It encompasses multiple interconnected data tables including campaigns, ad creatives, user demographics, and granular interaction events.

Business Context

The dataset enables end-to-end analysis of digital advertising effectiveness across Meta's ecosystem, supporting data-driven decision-making for marketing teams, analysts, and business stakeholders.

Use of Each Table

Table	Description	Primary Use
ad_events	Stores event-level records such as impressions, clicks, and purchases. It serves as the main fact table.	Forms the basis for KPI calculation and ad interaction analysis.
ads	Contains information about each ad creative, including campaign linkage and targeting details.	Used to analyze performance by platform, ad type, and creative format.
campaigns	Contains campaign-level strategy, budget, and timeframe details.	Used to evaluate campaign performance and calculate KPIs such as CPC, CPM, and ROAS.
users	Stores user demographic and interest data related to ad interactions.	Used for audience segmentation and to assess targeting effectiveness.

Table and Field Details

Table 1: ad_events

Purpose: Captures every interaction or event between a user and an ad.

Field	Description	Use in Analysis
event_id	Unique identifier for each event	Serves as the primary key and ensures each interaction is uniquely tracked
ad_id	Connects each event to a specific ad	Enables linking with the ads table for creative, platform, and format analysis
user_id	Connects each event to a specific user	Supports audience-level analysis by linking with demographic and interest data
timestamp	Records the exact date and time of the event	Helps build time-based analysis such as daily, weekly, and monthly trends
day_of_week	Indicates the weekday of the event	Useful for comparing weekday and weekend performance patterns
time_of_day	Classifies the event into a time segment	Helps identify when audience engagement is highest
event_type	Defines the type of interaction, such as impression, click, or purchase	Supports funnel tracking and KPI measurement across user actions

Usage: Forms the core analytical base for measuring campaign KPIs, including Impressions, Clicks, CTR, Conversion Rate, and ROAS.

Table 2: ads

Purpose: Defines ad-level metadata.

Field	Description	Use in Analysis
ad_id	Unique identifier for each ad	Serves as the primary key and connects ad-level details with event data
campaign_id	Links each ad to its respective campaign	Enables campaign-level performance and budget analysis
ad_platform	Identifies the platform where the ad is published, such as Facebook or Instagram	Supports comparison of performance across advertising platforms
ad_type	Specifies the creative format, such as image, video, carousel, or story	Helps evaluate which creative type delivers stronger results
target_gender	Indicates the gender segment targeted by the ad	Used to assess targeting accuracy and response by gender
target_age_group	Defines the age bracket targeted by the ad	Supports comparison between intended audience and actual engagement
target_interests	Captures the interest categories targeted by the ad	Helps evaluate alignment between targeting strategy and user behavior

Usage: Helps identify which platform and ad format perform best, and whether targeting aligns with actual audience engagement.

Table 3: campaigns

Purpose: Contains campaign-level information such as budget, duration, and strategy.

Field	Description	Use in Analysis
campaign_id	Unique identifier for each campaign	Serves as the primary key and links campaign data with ads
name	Name of the campaign	Used for reporting, filtering, and dashboard navigation
start_date	Date when the campaign begins	Helps track campaign launch and active periods
end_date	Date when the campaign ends	Supports campaign duration and completion analysis
duration_days	Derived field showing total campaign length	Used to compare campaign pacing and performance over time
total_budget	Total budget allocated to the campaign	Forms the basis for budget tracking and ROI-related KPI analysis

Usage: Enables budget tracking, campaign pacing analysis, and ROI evaluation.

Table 4: users

Purpose: Stores demographic and interest details of users who engage with ads.

Field	Description	Use in Analysis
user_id	Unique identifier for each user	Serves as the primary key and links user data with ad interaction events
user_gender	Gender of the user	Supports gender-based performance and audience response analysis
user_age	Age of the user	Forms the basis for detailed audience segmentation
age_group	Grouped age category such as 18–24 or 25–34	Helps compare engagement patterns across age segments
country	Country where the user is located	Used for country-level reach and performance analysis
location	More specific user location such as city or state	Supports geo-targeting and regional performance evaluation
interests	Interest categories associated with the user	Helps assess alignment between audience interests and ad targeting

Usage: Helps evaluate audience targeting accuracy by comparing intended target segments with actual user engagement.